a taxonomy of segregation measures

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There are a lot of segregation measures…

- Exposure indices: the average composition of the environments of members of one group
  - E.g., "the average proportion of non-immigrants in the neighborhood of the average immigrant"
- (Un)evenness indices: measures of how unevenly groups are distributed among contexts
  - Population-Relative
    - Typically range from 0-1, 1=maximal possible unevenness given overall population composition; 0=perfect evenness
  - Absolute
    - Degree of unevenness, not relative to overall population composition
  - Local
    - Measure of local composition/diversity relative to overall population composition

There are a lot of segregation measures…

- Binary indices: degree of segregation between two mutually-exclusive groups; e.g.:
  - Immigrant/non-immigrant
  - Male/Female
- Multigroup indices: degree of segregation between two+ mutually-exclusive, unordered groups; e.g.:
  - Muslim/Christian/Hindu/Buddhist/Other
  - White/Black/Asian/Indigenous
- Ordinal indices: degree of segregation between two+ mutually-exclusive, ordered groups; e.g.:
  - HS/Associate/Bachelor/Master/PhD
  - Manual Labor/Service/Skilled/Professional
- Continuous or rank-ordered indices: degree of segregation among individuals who vary by a continuous measure; e.g.:
  - Income
  - Academic skill

There are a lot of segregation measures…

- Aspatial indices: indices that measure segregation among contexts in which spatial location is irrelevant; e.g.:
  - Schools
  - Churches
- Spatial indices: indices that account for spatial proximity of contexts in measuring segregation; e.g.:
  - Neighborhoods
  - Proximity may be measured by Euclidean distance, travel time, activity space overlap, etc.
- Network indices: indices that account for network proximity or influence in measuring segregation; e.g.:
  - Friendship networks
  - Disciplinary citation patterns

A general framework of segregation measures

- Index persons by i and j
- Let x measure a group characteristic (may be binary, categorical, ordinal, continuous).
- Let ϕ_{ij} measure the *proximity* of j to i or the *influence* of j on i. (note that ϕ need not be symmetric).
  - Note that ϕ can measure spatial or network proximity
  - ϕ might be defined by a theoretical model of how individuals influence others, of how knowledge flows, etc.

A general framework of segregation measures

- Let $\rho_i(x)$ indicate the exposure density function of x in the local environment of i (the proportion of i's local environment made up of individuals in group x

$$\rho_i(x) = \frac{\sum_j \phi_{ij}(x_j = x)}{\sum_j \phi_{ij}}$$

- Let $\rho(x)$ indicate the density function of x in the population

$$\rho(x) = \frac{1}{N}\sum_j (x_j = x)$$

Examples of proximity functions

- Aspatial

$$\phi_{ij} = \begin{cases} 1, & i,j \; in \; same \; school \\ 0, & i,j \; in \; different \; school \end{cases}$$

- Spatial
  - ϕ_{ij} is declining function of Euclidean distance, or travel time, or an empirical activity-space derived metric
- Network
  - ϕ_{ij} is a measure of network proximity (e.g., declining function of degree of separation; probability that a random walk starting at j will reach i within some number of steps, etc).
- **Key insight**: "aspatial" measures are simply a special case of spatial measures.

A general framework of segregation measures

- Let d_i measure the diversity of the local environment of person i (this is a function of ρ_i); d_i is a measure of the variation in x within the local environment of person i

$$d_i = \int_x f\big(\rho_i(x)\big)dx$$

- Let D measure the diversity of the population

$$D = \int_x f\big(\rho(x)\big)dx$$

Examples of diversity functions

- Binary diversity: $x \in \{0,1\}$; note that $\rho_i(0) = 1 - \rho_i(1)$

$$I_i = \rho_i(0) \cdot \rho_i(1)$$
$$E_i = -\rho_i(0) \ln\big(\rho_i(0)\big) - \rho_i(1) \ln\big(\rho_i(1)\big)$$

- Multi-group diversity:

$$I_i = \sum_k \rho_i(x_k)[1 - \rho_i(x_k)]$$
$$E_i = -\sum_k \rho_i(x_k) \ln\big(\rho_i(x_k)\big)$$

Examples of diversity functions

- Ordinal diversity: define $c_k = I[x \leq x_k]$

$$I_i = \frac{1}{K-1} \sum_{k=1}^{K-1} \rho_i(c_k)[1 - \rho_i(c_k)]$$

$$E_i = \frac{-1}{K-1} \sum_{k=1}^{K-1} \rho_i(c_k) \ln\big(\rho_i(c_k)\big)$$

Examples of diversity functions

- Rank-order/continuous diversity: define $c_q = I[x \leq q], q \in [0,1]$

$$I_i = \int_0^1 \rho_i(c_q)\left[1 - \rho_i(c_q)\right]dq$$

$$E_i = -\int_0^1 \rho_i(c_q)\ln\left(\rho_i(c_q)\right)dq$$

A general framework of segregation measures

- Absolute Segregation: the difference between population diversity and the average diversity of individuals' local environments:

$$\Delta = \frac{1}{N}\sum_{i}(D - d_i) = D - \overline{D}$$

- Relative Segregation: absolute segregation divided by population diversity

$$S = \frac{\Delta}{D}$$

- Local Segregation: local diversity relative to population diversity

$$L_i = \frac{d_i}{D}$$

A general framework of segregation measures

- Relative Segregation: absolute segregation divided by population diversity

$$S = \frac{\Delta}{D}$$

- The relative diversity indices (V) and the information theory indices (H) have this form
- Two common indices do not have this form:
  - The Dissimilarity Index
  - The Gini Index of Segregation

A general framework of segregation measures

- Absolute Segregation: the difference between population diversity and the average diversity of individuals' local environments:

$$\Delta = \frac{1}{N}\sum_{i}(D - d_i) = D - \overline{D}$$

- Mutual information theory index has this form

Suggestions for writing general segregation code

- Four key inputs:
  - category type (binary, multigroup, ordinal, continuous)
  - proximity function
  - diversity function
  - type of measure (relative or absolute)
- Special cases:
  - Dissimilarity and gini indices
  - Exposure measures
  - Local measures